C PROGRAMMING LAB

Experiments 1–15 • Separate program listings

Note: These programs are prepared from the experiment topics in the lab manual. Compare with your teacher's required format before submitting.

Experiment 1: 01 basic input output

```
#include &lt;stdio.h&gt;
int main(void) {
    int a; float b; double c; char ch; char name[50];
    printf("Enter an integer: "); scanf("%d", &amp;a);
    printf("Enter a float value: "); scanf("%f", &amp;b);
    printf("Enter a double value: "); scanf("%lf", &amp;c);
    printf("Enter a single character: "); scanf(" %c", &amp;ch);
    printf("Enter a string (name): "); scanf("%49s", name);
    printf("\n--- Output ---\n");
    printf("Integer entered : %d\n", a);
    printf("Float entered : %.2f\n", b);
    printf("Double entered : %.2lf\n", c);
    printf("Character entered : %c\n", ch);
    printf("String entered : %s\n", name);
    return 0;
}
```

Experiment 2: 02 area of circle

```
#include &lt;stdio.h&gt;
#define PI 3.14159
int main(void) {
    float radius, area;
    printf("Enter the radius of the circle: ");
    scanf("%f", &amp;radius);
    area = PI * radius * radius;
    printf("Area of the circle = %.2f\n", area);
    return 0;
}
```

Experiment 3: 03 average of three

```
#include &lt;stdio.h&gt;
int main(void) {
    float a, b, c, average;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &amp;a, &amp;b, &amp;c);
    average = (a + b + c) / 3;
    printf("Average of the three numbers = %.2f\n", average);
    return 0;
}
```

Experiment 4: 04 swap using temporary

```
#include &lt;stdio.h&gt;
int main(void) {
    int a, b, temp;
    printf("Enter two numbers: ");
    scanf("%d %d", &amp;a, &amp;b);
    printf("Before swapping: a = %d, b = %d\n", a, b);
    temp = a; a = b; b = temp;
    printf("After swapping: a = %d, b = %d\n", a, b);
    return 0;
}
```

Experiment 5: 05 single and multiline comments

```
#include &lt;stdio.h&gt;
int main(void) {
    // This is a single-line comment.
    /* This is a multi-line comment.
       Comments explain code and are ignored by the compiler. */
    int num = 10; // Initialize a variable.
    printf("The value of num is: %d\n", num);
    return 0;
}
```

Experiment 6: 06 operator precedence

```
#include &lt;stdio.h&gt;
int main(void) {
    int a = 5, b = 10, c = 2, result1, result2;
    result1 = a + b * c;
    result2 = (a + b) * c;
    printf("a + b * c      = %d\n", result1);
    printf("(a + b) * c    = %d\n", result2);
    int x = 5, y = 10, z = 15;
    int logicResult = x &lt; y &amp;&amp; y &lt; z;
    printf("x &lt; y &amp;&amp; y &lt; z = %d\n", logicResult);
    return 0;
}
```

Experiment 7: 07 even or odd

```
#include &lt;stdio.h&gt;
int main(void) {
    int number;
    printf("Enter a number: "); scanf("%d", &amp;number);
    if (number % 2 == 0) printf("%d is Even\n", number);
    else printf("%d is Odd\n", number);
    return 0;
}
```

Experiment 8: 08 positive negative zero

```
#include &lt;stdio.h&gt;
int main(void) {
    int number;
    printf("Enter a number: "); scanf("%d", &amp;number);
    if (number &gt; 0) printf("%d is Positive\n", number);
    else if (number &lt; 0) printf("%d is Negative\n", number);
    else printf("The number is Zero\n");
    return 0;
}
```

Experiment 9: 09 menu driven calculator

```
#include &lt;stdio.h&gt;
int main(void) {
    int choice; double a, b;
    do {
        printf("\n----- Menu-Driven Calculator -----\n");
        printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n0. Exit\n");
        printf("Enter your choice: "); scanf("%d", &amp;choice);
        if (choice &gt;= 1 &amp;&amp; choice &lt;= 4) {
            printf("Enter two numbers: "); scanf("%lf %lf", &amp;a, &amp;b);
            switch (choice) {
                case 1: printf("Result = %.2lf\n", a + b); break;
                case 2: printf("Result = %.2lf\n", a - b); break;
                case 3: printf("Result = %.2lf\n", a * b); break;
                case 4: if (b != 0) printf("Result = %.2lf\n", a / b); else printf("Cannot divide
by zero.\n"); break;
            }
        } else if (choice != 0) printf("Invalid choice. Try again.\n");
    } while (choice != 0);
    printf("Calculator closed.\n");
    return 0;
}
```

Experiment 10: 10 multiplication table three loops

```
#include &lt;stdio.h&gt;
int main(void) {
    int n, i;
    printf("Enter a number: "); scanf("%d", &amp;n);
    printf("\nMultiplication Table using while loop:\n");
    i = 1; while (i &lt;= 10) { printf("%d x %d = %d\n", n, i, n * i); i++; }
    printf("\nMultiplication Table using do-while loop:\n");
    i = 1; do { printf("%d x %d = %d\n", n, i, n * i); i++; } while (i &lt;= 10);
    printf("\nMultiplication Table using for loop:\n");
    for (i = 1; i &lt;= 10; i++) printf("%d x %d = %d\n", n, i, n * i);
    return 0;
}
```

Experiment 11: 11 integer float character arrays

```
#include &lt;stdio.h&gt;
int main(void) {
    int integers[5], i; float floats[5]; char chars[5];
    printf("Enter 5 integer values:\n");
    for (i = 0; i &lt; 5; i++) scanf("%d", &amp;integers[i]);
    printf("Enter 5 float values:\n");
    for (i = 0; i &lt; 5; i++) scanf("%f", &amp;floats[i]);
    printf("Enter 5 characters:\n");
    for (i = 0; i &lt; 5; i++) scanf(" %c", &amp;chars[i]);
    printf("Integer array: "); for (i = 0; i &lt; 5; i++) printf("%d ", integers[i]);
    printf("\nFloat array: "); for (i = 0; i &lt; 5; i++) printf("%.2f ", floats[i]);
    printf("\nCharacter array: "); for (i = 0; i &lt; 5; i++) printf("%c ", chars[i]);
    printf("\n"); return 0;
}
```

Experiment 12: 12 sum of array elements

```
#include &lt;stdio.h&gt;
int main(void) {
    int a[100], n, i, sum = 0;
    printf("Enter number of elements (1-100): "); scanf("%d", &amp;n);
    if (n &lt; 1 || n &gt; 100) { printf("Invalid size.\n"); return 1; }
    printf("Enter %d elements: ", n);
    for (i = 0; i &lt; n; i++) { scanf("%d", &amp;a[i]); sum += a[i]; }
    printf("Sum of array elements = %d\n", sum);
    return 0;
}
```

Experiment 13: 13 matrix addition

```
#include &lt;stdio.h&gt;
int main(void) {
    int a[10][10], b[10][10], sum[10][10], r, c, i, j;
    printf("Enter rows and columns (1-10): "); scanf("%d %d", &amp;r, &amp;c);
    if (r &lt; 1 || r &gt; 10 || c &lt; 1 || c &gt; 10) { printf("Invalid matrix size.\n"); return
1; }
    printf("Enter elements of first matrix:\n");
    for (i = 0; i &lt; r; i++) for (j = 0; j &lt; c; j++) scanf("%d", &amp;a[i][j]);
    printf("Enter elements of second matrix:\n");
    for (i = 0; i &lt; r; i++) for (j = 0; j &lt; c; j++) scanf("%d", &amp;b[i][j]);
    for (i = 0; i &lt; r; i++) for (j = 0; j &lt; c; j++) sum[i][j] = a[i][j] + b[i][j];
    printf("Sum of matrices:\n");
    for (i = 0; i &lt; r; i++) { for (j = 0; j &lt; c; j++) printf("%d ", sum[i][j]); printf("\n");
}
    return 0;
}
```

Experiment 14: 14 factorial using function

```
#include &lt;stdio.h&gt;
unsigned long long factorial(int n) {
    unsigned long long result = 1;
    for (int i = 1; i &lt;= n; i++) result *= (unsigned long long)i;
    return result;
}
int main(void) {
    int n;
    printf("Enter a non-negative integer (0-20): "); scanf("%d", &amp;n);
    if (n &lt; 0 || n &gt; 20) { printf("Please enter a number from 0 to 20.\n"); return 1; }
    printf("Factorial of %d = %llu\n", n, factorial(n));
    return 0;
}
```

Experiment 15: 15 triangle area four function types

```
#include &lt;stdio.h&gt;
#define HALF 0.5
/* 1. No arguments, no return value */
void area1(void) {
    float b, h; printf("\n[No arguments, no return value]\nEnter base and height: ");
    scanf("%f %f", &amp;b, &amp;h); printf("Area = %.2f\n", HALF * b * h);
}
/* 2. Arguments, no return value */
void area2(float b, float h) { printf("Area = %.2f\n", HALF * b * h); }
/* 3. No arguments, returns a value */
float area3(void) {
    float b, h; printf("\n[No arguments, returns value]\nEnter base and height: ");
    scanf("%f %f", &amp;b, &amp;h); return HALF * b * h;
}
/* 4. Arguments, returns a value */
float area4(float b, float h) { return HALF * b * h; }
int main(void) {
    float b, h;
    area1();
    printf("\n[Arguments, no return value]\nEnter base and height: "); scanf("%f %f", &amp;b, &amp;
h); area2(b, h);
    printf("\nArea = %.2f\n", area3());
    printf("\n[Arguments, returns value]\nEnter base and height: "); scanf("%f %f", &amp;b, &amp;h);
 printf("Area = %.2f\n", area4(b, h));
    return 0;
}
```